



**Facultad de
Ciencias**
UNAM

Characterization of plasmonic disordered metasurfaces for biosensing applications under realistic conditions

PhD Protocol Summary

Jonathan Alexis Urrutia Anguiano

Facultad de Ciencias
UNAM

jaurrutia.95@ciencias.unam.mx

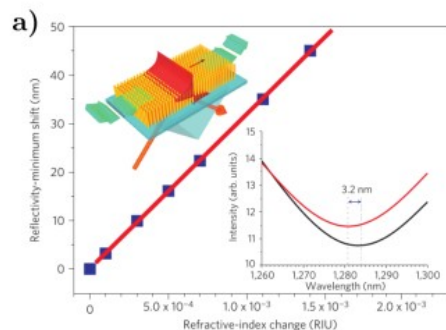


Committee & Contents

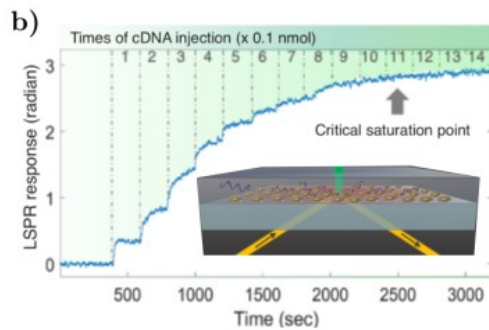
- **Principal Tutor**
 - Dr. Alejandro Reyes Coronado
Facultad de Ciencias, UNAM
- **Committee Members**
 - Dr. Rubén García Ramos
Instituto Nacional de Astrofísica, Óptica y Electrónica (INAOE)
 - Dr. W. Luis Mochán Backal
Instituto de Ciencias Físicas, UNAM

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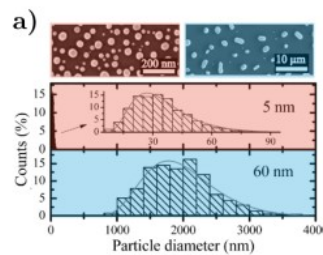
Background and Objectives



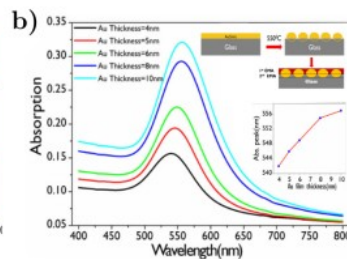
A. V. Kabashin et al.
Nature Mater.
8(11):867-871, 2009.



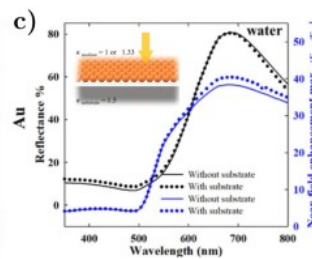
G. Qiu et al.
ACS Nano.
14(5):5268-5277, 2020.



D. Wang et al.
Beilstein J. Nanotechnol.
8:318-326, 2011.



R.S. Moirangthem et al.
Biomed. Opt. Express.
3(5):899, 2012.



R. Borah et al.
Sci. Rep.
12(1):15738, 2022.

Theoretical-Experimental Collaboration between FC-UNAM & INAOE

- Disordered metasurfaces aimed for Biosensing Applications
 - Implementation of Analytical and Numerical Techniques
 - Multiple Scattering Theories
 - Finite Element Method
 - Parameter Fitting Method: Descent Gradient
 - Quantification of Multiple Scattering and Substrate Effects
 - Design proposal and spectral analysis of biosensing-aimed metasurfaces

Theoretical Frameworks and Measurement Techniques

Coherent Scattering Theory

$$r_{\text{CSM}}^{(j)}(\theta) = \frac{-\beta_+^{(j)}(\theta)}{1 + \beta_F + \frac{1}{4}[\beta_F^2 - \beta_+^{(j)}(\theta)\beta_-^{(j)}(\theta)]},$$

$$\beta_F(\theta) = \eta \int_0^\infty \rho(a) S(0) da \quad \text{and} \quad \beta_\pm^{(j)}(\theta) = \eta \int_0^\infty \rho(a) S_j(\pi - 2\theta) \exp(\pm 2ik_z^{\text{inc}} a) da,$$

O. vázquez-Estrada et al.
J. Opt. Soc. Am. A.
 31(4):745, 2014.

Thin Film Theory

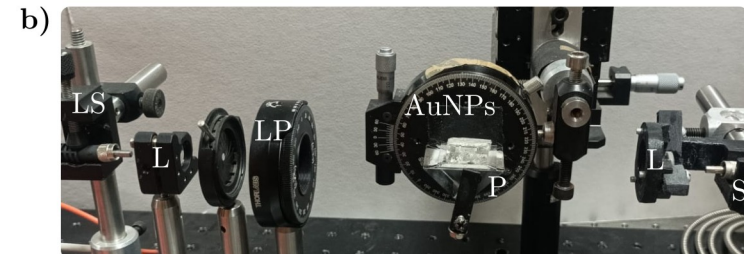
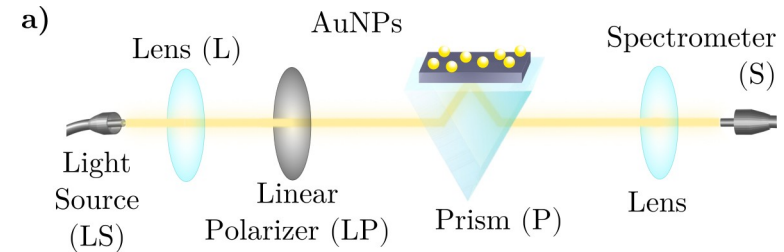
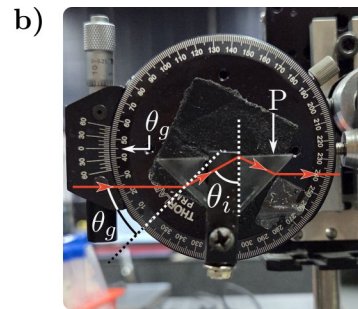
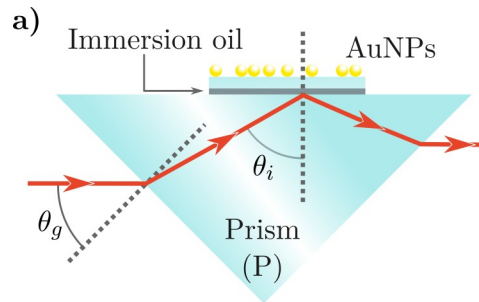
$$r_{\text{TFT}}^{(p)}(\theta_i) = \frac{\kappa_-(\omega) - i(\omega/c)\gamma \cos \theta_i \cos \theta_t + i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2 \theta_i}{\kappa_+(\omega) - i(\omega/c)\gamma \cos \theta_i \cos \theta_t - i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2 \theta_i},$$

$$t_{\text{TFT}}^{(p)}(\theta_i) = \frac{n_i \cos \theta_i (1 - (\omega/2c)^2 \varepsilon_i \gamma \beta \sin^2 \theta_i)}{\kappa_+(\omega) - i(\omega/c)\gamma \cos \theta_i \cos \theta_t - i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2 \theta_i},$$

$$\kappa_\pm(\omega) = \left(n_t \cos \theta_i \pm n_i \cos \theta_t \right) \left[1 - \left(\frac{\omega}{2c} \right)^2 \varepsilon_i \gamma \beta \sin^2 \theta_i \right].$$

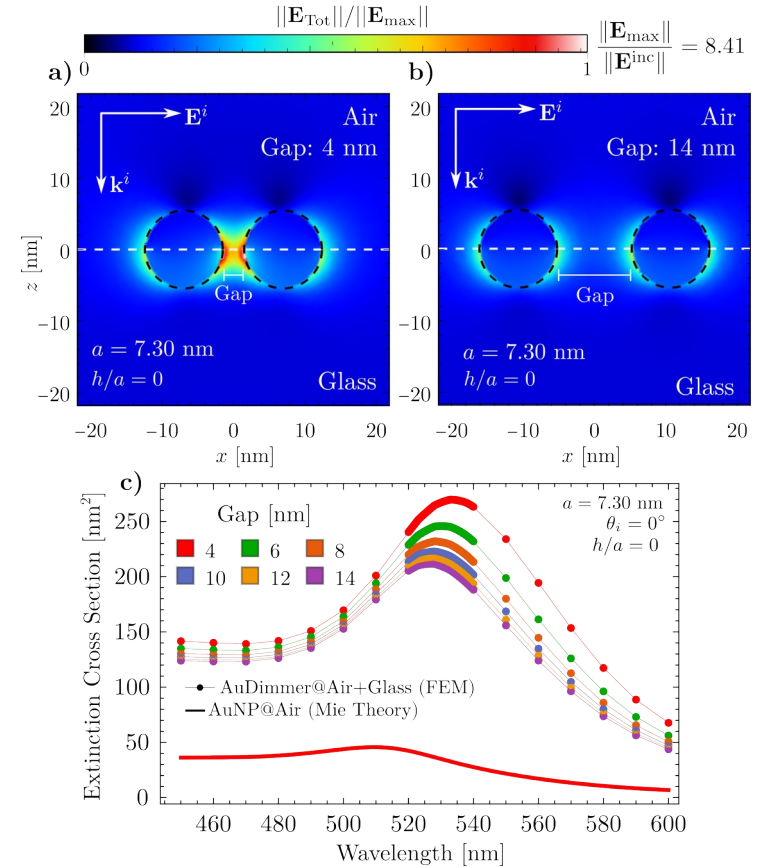
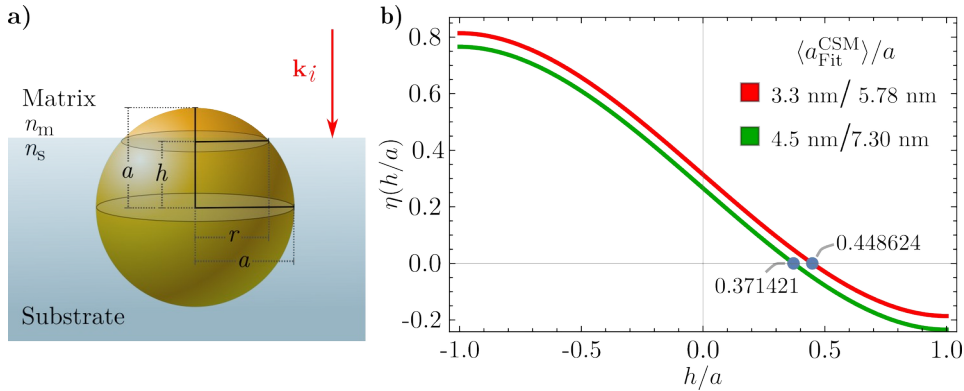
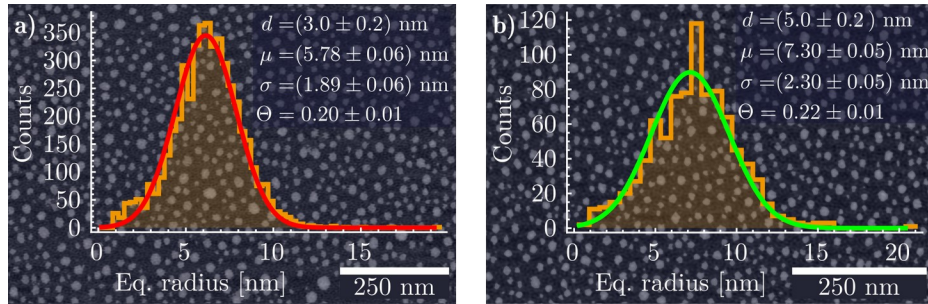
D. Bedeaux and J. Vlieger.
 Optical properties of surfaces.
 Imperial College Press, London 2nd Ed. , 2004

Reflectivity Spectra Setup at INAOE



Progress in Non-destructive Characterization

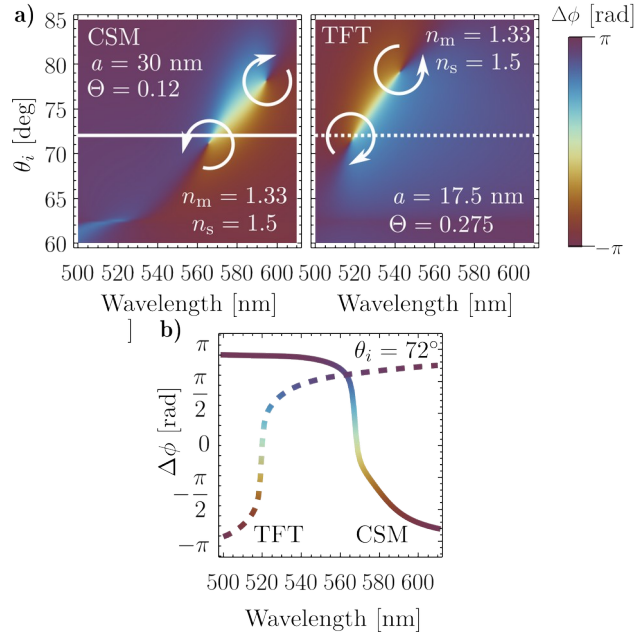
Partial embedding of the metasurface



Progress in Optical Response and Sensing Mechanisms of Metasurfaces

Topological Darkness Phenomenon

G. Ermolaev et al.
Nat. Commun.
13(1):2049, 2022.



Phase-based sensing schema for refractive index change

